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标题	Preparation of Acidic 5-Hydroxy-1,2,3-triazoles via the Cycloaddition of Aryl Azides with β-Ketoesters
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摘要（原文节选）

Herein, a high-yielding cycloaddition reaction of β-ketoesters and azides to provide 1,2,3-triazoles is described. The reactions employing 2-unsubstituted β-ketoesters were found to provide 5-methyl-1,2,3-triazoles, whereas 2-alkyl-substituted β-ketoesters provided 5-hydroxy-1,2,3-triazoles (shown to be relatively acidic) in high yields and as single regioisomers. Several novel compounds were reported and characterized including long-chain 5-hydroxy-1,2,3-triazoles potentially bioisosteric to hydroxamic acids. ⋯

数据表 · Table 1（4 行，含数值行 4 行）

Cycloaddition of β-Ketoester 1a with Aryl Azides 2a

entry	keto ester	aryl azide	Py	product	catalyst	solvent	yield (%)b
1	1a	2a	4-pyridyl	3a		MeCN	80
2	1a	2b	3-pyridyl	3b		MeCN	72
3	1a	2c	2-pyridyl	3c		MeCN	0
4	1a	2c	2-pyridyl	3c	Cu(OTf)2 · C6H5 CH3	DMSO	50

脚注（逐字）：aReaction conditions are as follows: 1a (1.2 equiv), 2a – c (1 equiv), DBU (1.2 equiv), and solvent (0.2 M).bIsolated yield.

正文对应句（逐字）：Our investigation began with the cycloaddition reaction of pyridyl azides 2a – c with ethyl acetoacetate 1a in the presence of organic soluble bases, from which 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU) gave the best yield. The reactions of pyridyl azides 2a and 2b with 1a furnished triazoles 3a and 3b

数据表 · Table 2（13 行，含数值行 13 行）

Cycloaddition of β-Ketoesters and Aryl Azides to Give 5-Hydroxytriazolesa

entry	ketoester	R	aryl azide	Ar	product	time (h)	yield (%)b
1	1a	H	4a	Ph	3d	18	87
2	1b	Me	4a	Ph	5a	6	80
3	1b	Me	4a	Ph	5a	18	98c
4	1b	Me	4b	4-BrC6H4	5b	18	87
5	1b	Me	4c	4-OMeC6H4	5c	18	62
6	1b	Me	4d	2-PhC6H4	5d	18	75
7	1b	Me	4e	1-naphthyl	5e	18	79
8	1b	Me	4f	4-NO2C6H4	5f	18	53
9	1b	Me	4g	2-NO2-4-MeC6H3	5g	18	56
10	1b	Me	4h	2-Br-C6H4	5h	18	n.r.d
11	1b	Me	4i	n-Bu	5i	18	n.r.d
12	1b	Me	4j	Bn	5j	18	n.r.d
13	1c	Bn	4a	Ph	5k	18	75

脚注（逐字）：aReaction conditions are as follows: 1b or 1c (1.2 equiv), 4a – j (1 equiv), and DBU (1.2 equiv) in MeCN (0.2 M) at 50 ° C.bIsolated yield.c1H-NMR spectroscopic yield.dNo reaction.

正文对应句（逐字）：Our investigation began with the cycloaddition reaction of pyridyl azides 2a – c with ethyl acetoacetate 1a in the presence of organic soluble bases, from which 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU) gave the best yield. The reactions of pyridyl azides 2a and 2b with 1a furnished triazoles 3a and 3b

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